

ONDA LIFE

**Neuro-Adaptive Digital Health Platform for Research &
Clinical Validation**

SCIENTIFIC COLLABORATION PROPOSAL • RADBOUDUMC & DONDEERS INSTITUTE

Scientific Vision: Bridging Technology and Cognitive Psychophysiology

ONDA Tech Core

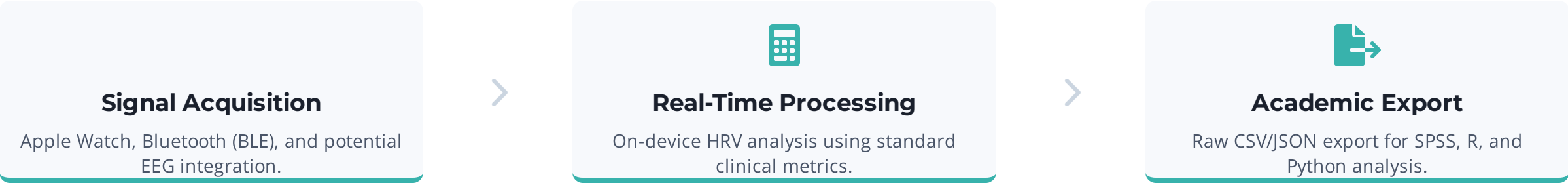
-  Real-time biofeedback engine for HRV and biometric integration.
-  High-fidelity data pipeline for raw signal logging and export.
-  Scalable mobile infrastructure (iOS/Android) for large-scale studies.



Donders Expertise

-  Fundamental research in cognitive control and stress mechanisms.
-  Gold-standard methodologies for attention and neuroplasticity.
-  Academic leadership in neuro-biological validation of interventions.

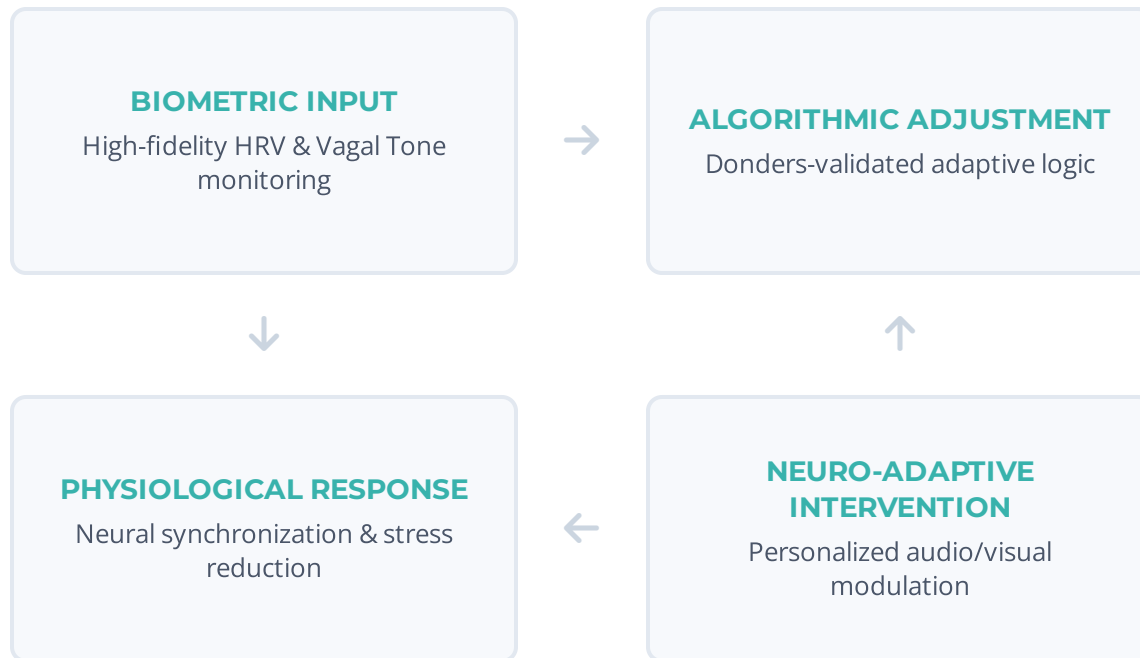
High-Fidelity Data Pipeline for Longitudinal Research



Metric Category	Specific Parameters Collected	Clinical Relevance
Time-Domain HRV	NN intervals, SDNN, rMSSD, pNN50	Vagal tone and parasympathetic activity
Frequency-Domain	HF/LF Power, LF/HF Ratio, Total Power	Sympathovagal balance and stress response
Contextual Data	Session type, time of day, user-reported mood	Correlation with physiological markers

"ONDA is designed not as a closed commercial application, but as an open-pipeline digital health platform capable of logging high-fidelity bio-signals for subsequent statistical analysis."

Co-Designing Closed-Loop Neuro-Adaptive Protocols



Collaboration Areas

-  **Protocol Validation:** Scientific proof of efficacy for closed-loop interventions.
-  **Neural Sync:** Optimizing audio frequencies for specific brain states.
-  **Vagal Toning:** Targeting the parasympathetic nervous system via biofeedback.

"We invite Radboudumc to co-design and neurobiologically validate the closed-loop intervention mechanisms."

Proposed Eurostars Joint Study: Clinical Validation Design

Core Hypothesis

"Daily 10-minute neuro-adaptive sessions will significantly improve vagal tone and reduce cortisol markers in professionals experiencing high levels of occupational burnout."

Study Design

-  **Population:** N=60-100 participants (high-stress corporate environment).
-  **Method:** Randomized Controlled Trial (RCT) with 12-week longitudinal follow-up.
-  **Timeline:** 2026-2027 (Eurostars Project Cycle).

Primary Outcomes

Physiological Markers

Assessment of HRV (rMSSD, HF Power) and salivary cortisol modulation.

Neuroplasticity Indicators

Evaluation of PFC activation and neural synchronization via Donders Institute protocols.

Psychological Scaling

Validated burnout scales (MBI) and cognitive performance metrics.

Academic Impact: Publications, Standards, and Scaling



Research Excellence

- ✓ Targeting Q1/Q2 journals: Frontiers in Neuroscience, Nature Digital Medicine.
- ✓ Joint authorship on fundamental neuro-adaptive mechanisms.
- ✓ Contribution to the global body of digital health evidence.



Industry Standards

- ✓ Establishing a de-facto standard for predictive burnout assessment.
- ✓ Open-standard HRV metrics for corporate wellness validation.
- ✓ Benchmarking neuro-adaptive interventions against traditional methods.



Practical Scaling

- ✓ Transition from clinical validation to large-scale B2B deployment.
- ✓ Scaling validated protocols across the European healthtech market.
- ✓ Long-term data partnership for ongoing longitudinal studies.

"Our goal is to transform fundamental neuroscientific insights into a validated digital tool that improves the functional state of millions."

Founding Team: Scientific Rigor Meets Technical Execution

Valentin

Co-founder & Scientific Advisor

PH.D. PHYSICS & NEUROSCIENCE

-  Expertise in biometric feedback loops and neural optimization mechanisms.
-  Oversees scientific methodology and clinical validation protocols.
-  Bridge between fundamental research and digital health application.

Yakiv

Founder & CEO

FULL-STACK EXECUTION

-  Lead developer responsible for ONDA's high-fidelity data pipeline.
-  1,871+ commits on the core codebase (iOS, Android, Supabase).
-  Architect of the open-pipeline system for academic data export.

ONDA LIFE

Advancing Digital Neuroscience Through Collaborative Innovation



Platform

onda-life.com



Research Inquiry

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Documentation

<https://onda-life.com/p/info>

Let's discuss how ONDA can support your research goals and clinical validation studies.